Supplementary Material

GCZ-RL: Geometry-Calibrated Residual Learning for Spinal Ligament Landmark Localization on Vertebral Surface Meshes

Section, equation, and table numbers without the prefix S refer to the main text. Every value in this document is regenerated by the scripts of the code release (`paper/make_paper_numbers.py`); the mapping from each table to its script and output file is given in `docs/PAPER_MAP.md`.

S1. Residual-regressor features

Each GCZ point $\mathbf{p}$ of ligament class L on a vertebra with vertex set $\mathcal{V}$ is described by the 36 features of Table S1. The notation is as follows. $\mathbf{v}_{\min}$ and $\mathbf{v}_{\max}$ are the componentwise minimum and maximum of $\mathcal{V}$, and $\mathbf{s} = \mathbf{v}_{\max} - \mathbf{v}_{\min}$ is the extent of the vertebra (components equal to zero are set to one). The normalized position is $\mathbf{p}_n = (\mathbf{p} - \mathbf{v}_{\min}) \oslash \mathbf{s} = (x_n, y_n, z_n)$, where $\oslash$ denotes componentwise division. $\mathcal{R}_L$ is the candidate region of class L used for projection (Section 3.4; if it contains fewer than two vertices, $\mathcal{V}$ is used), $\mathbf{c}_R$ its centroid, and $\mathbf{r}_{\min}$, $\mathbf{r}_{\max}$ its componentwise minimum and maximum. $[T_y^-, T_y^+]$ is the calibrated y_n band of class L. $\bar{\mathbf{b}}_L$ is the mean normalized expert landmark centroid of class L over the training vertebrae, and $\mathbf{b} = \mathbf{v}_{\min} + \bar{\mathbf{b}}_L \odot \mathbf{s}$ is its position on the current vertebra. $\mathcal{N}$ denotes the 30 vertices of $\mathcal{V}$ nearest to $\mathbf{p}$. All features are computed from the mesh, the GCZ point, and quantities learned from the training vertebrae; no annotation of the vertebra being corrected is used. Distances are in millimeters.

The features are produced by `point_features` and `_zone_features` in `code/ligament/gczrl.py` of the code release. Features 5 and 6 repeat features 2 and 3; they are listed because they are part of the implemented feature vector with which all reported results were obtained. For mirrored training samples (Section 3.7), the features are computed for the mirrored point on the original mesh.

Table S1: Definition of the 36 residual-regressor features.

#	Feature	Definition	Source
Position of the GCZ point			
1–3	Normalized position	x_n, y_n, z_n	Mesh bounding box
4	Distance from mid-sagittal plane	$\lvert x_n - 0.5 \rvert$	Mesh bounding box
5–6	Normalized position (repeat)	y_n, z_n	Mesh bounding box
Relation to the candidate region			
7	Distance to region centroid	$\lVert \mathbf{p} - \mathbf{c}_R \rVert_2$	Candidate region
8	Distance to region box boundary	$\min\big(\min_k(p_k - r_{\min,k}),\ \min_k(r_{\max,k} - p_k)\big)$; positive inside the bounding box of $\mathcal{R}_L$	Candidate region
9	Relative position in y_n band	$(y_n - T_y^-)/(T_y^+ - T_y^-)$; 0.5 if the band is degenerate	Calibrated thresholds
10–12	Normalized region centroid	$(\mathbf{c}_R - \mathbf{v}_{\min}) \oslash \mathbf{s}$	Candidate region
13–15	Normalized region spread	componentwise standard deviation of $\mathcal{R}_L$, divided by $\mathbf{s}$	Candidate region
16	Region size	number of vertices in $\mathcal{R}_L$	Candidate region
17–19	Region shape	singular values $\sigma_1, \sigma_2, \sigma_3$ of the centered vertices of $\mathcal{R}_L$, divided by $\sigma_1 + \sigma_2 + \sigma_3$	Candidate region
Anatomical context			
20	Vertebral level	index of the level in the order C1–C7, T1–T13, L1–L5, divided by 24	Level label
21–27	Ligament class	one-hot encoding of ALL, PLL, LF, ISL, SSL, ITL, CL	Class label
Local surface			
28–30	Surface normal	right singular vector of the smallest singular value of the centered set $\mathcal{N}$ (unit length)	Mesh, 30 nearest vertices
31	Surface variation	$\sigma_3/(\sigma_1 + \sigma_2 + \sigma_3)$ of the centered set $\mathcal{N}$	Mesh, 30 nearest vertices

#	Feature	Definition	Source
32	Signed lateral position	$x_n - 0.5$	Mesh bounding box
Offset from the training prior			
33	Distance to class prior	$\|\mathbf{p} - \mathbf{b}\|_2$	Training annotations (class mean)
34–36	Direction to class prior	$(\mathbf{b} - \mathbf{p}) \oslash \mathbf{s}$	Training annotations (class mean)

S2. Candidate-region construction

For class L, three normalized quantities are calibrated: y_n, $|x_n - 0.5|$, and z_n. For each, the mean μ_L and sample standard deviation σ_L of the expert landmarks of the training vertebrae, normalized within their own vertebra, give the band $[\mu_L - k_L\sigma_L,\ \mu_L + k_L\sigma_L]$ with $k_L = 2$, clipped to $[0, 1]$ (Eq. 3). The candidate region is the set of mesh vertices that lie inside all three bands. If fewer than 64 vertices remain, the z_n band is dropped and the region is formed from the y_n and $|x_n - 0.5|$ bands; in the GCZ selection step, if fewer than 64 vertices still remain, only the y_n band is used. If no vertex satisfies the bands, the whole mesh is used. These rules are implemented in `build_thresholds` and `candidate_vertices` of `code/ligament/gcz.py`.

S3. Point accounting and matching-independent metrics

The following statistics refer to the primary, patient-level leave-one-subject-out evaluation (14 lumbar vertebrae, 98 vertebra–class cases) and are computed from the saved predictions and annotations of the code release; no model was retrained. The expert annotations are dense point sets along each attachment footprint, whereas GCZ and GCZ-RL output ten landmarks per class (one for SSL and two for ITL) and SLD a class-dependent number. One-to-one Hungarian matching within each vertebra–class case pairs $\min(|P|, |G|)$ points; in every case the predicted set was the smaller one, so every predicted landmark was paired and no predicted landmark was left unpaired. No method had a missing output for any case.

Table S2: Point accounting of the primary evaluation. Ref.: number of expert reference points. Pred.: number of predicted landmarks, all of which were matched. Cov.: matched reference points as a percentage of all reference points. Unpaired reference points are Ref. minus Pred.

		GCZ		GCZ-RL		SLD	
	Ref.	Pred.	Cov.%	Pred.	Cov.%	Pred.	Cov.%
By class							
ALL	368	140	38.0	140	38.0	196	53.3
PLL	239	140	58.6	140	58.6	196	82.0
LF	349	140	40.1	140	40.1	132	37.8
ISL	330	140	42.4	140	42.4	196	59.4
SSL	14	14	100.0	14	100.0	14	100.0
ITL	28	28	100.0	28	100.0	28	100.0
CL	837	140	16.7	140	16.7	176	21.0
By held-out subject							
S01	1020	265	26.0	265	26.0	337	33.0
S02	564	212	37.6	212	37.6	264	46.8
S03	581	265	45.6	265	45.6	337	58.0
All	2165	742	34.3	742	34.3	938	43.3

Because matched-pair metrics do not penalize reference points without a prediction, Table S3 reports metrics that use every predicted and every reference point and do not depend on matching. The closest-point distance from predictions to references (P→R) is the mean, over predicted landmarks, of the distance to the nearest reference point; it is the error definition of Blomenkamp et al. (2026) and is never larger than the matched MAE. The closest-point distance from references to predictions (R→P) is the mean, over reference points, of the distance to the nearest prediction; it penalizes parts of the footprint that no prediction covers. The symmetric distance is the mean of the two, and the last block is the percentage of all reference points whose nearest prediction lies within 5 mm.

Table S3: Matching-independent metrics of the primary evaluation, averaged over the vertebra–class cases of each class; the last row averages over all 98 cases (equal to the macro average, since every class has 14 cases). Lowest distance and highest percentage per row and block in bold.

	P→R closest point (mm)			R→P closest point (mm)			Symmetric (mm)			Reference points within 5 mm (%)		
Class	GCZ	GCZ-RL	SLD	GCZ	GCZ-RL	SLD	GCZ	GCZ-RL	SLD	GCZ	GCZ-RL	SLD
ALL	2.605	2.382	**1.521**	3.989	3.798	**3.175**	3.297	3.090	**2.348**	70.5	75.4	**78.4**
PLL	1.846	1.846	**1.177**	2.329	2.329	**1.969**	2.087	2.087	**1.573**	**97.2**	**97.2**	92.3
LF	1.918	**1.912**	4.015	3.334	**3.323**	9.829	2.626	**2.618**	6.922	**80.8**	**80.8**	34.3
ISL	2.262	2.173	**1.367**	3.706	3.621	**3.143**	2.984	2.897	**2.255**	77.0	77.3	**82.1**
SSL	2.917	2.692	**1.968**	2.917	2.692	**1.968**	2.917	2.692	**1.968**	78.6	78.6	**92.9**
ITL	6.262	1.796	**0.873**	6.262	1.796	**0.873**	6.262	1.796	**0.873**	53.6	92.9	**100.0**
CL	2.302	2.180	**1.591**	5.631	**5.522**	7.664	3.966	**3.851**	4.628	46.8	46.9	**63.4**
Mean over cases	2.873	2.140	**1.787**	4.024	**3.297**	4.089	3.448	**2.719**	2.938	72.1	**78.4**	77.6

S4. Candidate-region feasibility

For each outer fold of the primary evaluation, the candidate regions of the held-out vertebrae were rebuilt from the training subjects exactly as in the method (Section S2), and each reference landmark of the held-out subject was assigned to the mesh vertex nearest to it. A landmark counts as inside if that vertex belongs to the region of its class. The GCZ selection step and the GCZ-RL projection step use the same bands but differ in their last fallback (Section S2); they produce identical regions for every class except ITL. By vertebral level, the percentage of reference landmarks inside their region was L1 87.5%, L2 96.9%, L3 95.1%, L4 89.5%, L5 76.2%.

Table S4: Candidate-region feasibility on the held-out vertebrae of the primary evaluation (14 vertebrae per class). Inside: reference landmarks whose nearest mesh vertex lies in the region of their class. Region size: median number of vertices (meshes have a median of 33,445 vertices). Fallback: cases in which fewer than 64 vertices satisfied all three bands. Side empty: ITL or CL cases in which the projection region had no vertex on one side of the mid-sagittal plane.

			Region size		Fallback		
Class	Ref.	Inside%	Selection	Projection	Selection	Projection	Side empty
ALL	368	96.7	6041	6041	0/14	0/14	0
PLL	239	83.7	2891	2891	0/14	0/14	0
LF	349	87.4	5254.5	5254.5	0/14	0/14	0
ISL	330	94.5	2764	2764	0/14	0/14	0
SSL	14	85.7	124.5	124.5	5/14	5/14	0
ITL	28	92.9	10990	34.5	14/14	14/14	2
CL	837	88.1	8467.5	8467.5	0/14	0/14	0
All	2165	90.0	–	–	19/98	19/98	2

The two side-empty cases are the two ITL vertebrae (S01 L5 and S02 L4) for which the corrected point keeps its GCZ position (Section 4.3).

S5. Implementation details and hyperparameters

Table S5 lists the settings of every component. All are fixed; none was tuned on held-out annotations except the per-class development configuration (Section 3.7), whose selection after inspection of held-out outputs is disclosed in the main text.

Table S5: Implementation settings.

Component	Setting
Candidate regions	$k_L = 2$; bands clipped to $[0, 1]$; fallback below 64 vertices (Section S2).
Atlas selection	ICP from every atlas vertebra, 1200 sampled atlas points, at most 40 iterations, tolerance 10^{-5}; residual = mean nearest-neighbor distance of 1200 sampled atlas points.
Full alignment	ICP, 5000 sampled points, at most 150 iterations, tolerance 10^{-6} (primary evaluation and development GCZ column) or 10^{-5} (development GCZ-RL input).

Component	Setting
GCZ selection	Snap radius 15 mm; left/right split for LF and ITL, four quadrants for CL; farthest-point sampling to ten landmarks. A fallback landmark template pooled from all training vertebrae is used only if the selected atlas lacks a class.
Residual targets	Hungarian pairing of GCZ and expert points; mirroring of LF, ISL, ITL, CL about the sagittal plane through the vertex centroid.
Correction bound	τ_L = mean + 2 SD of training residual magnitudes of class L (including mirrored samples).
Extra Trees ET-1	220 trees, maximum depth 18, minimum leaf size 1, 85% of features per split, no bootstrap, seed 8.
Extra Trees ET-2	520 trees, unlimited depth, minimum leaf size 1, all features, no bootstrap, seed 91.
Extra Trees ET-3	520 trees, maximum depth 13, minimum leaf size 2, 75% of features, no bootstrap, seed 92.
Extra Trees ET-4	520 trees, maximum depth 18, minimum leaf size 1, 85% of features, bootstrap, seed 93.
Other regressors (Table 3)	Ridge ($\lambda = 25$, standardized features); k-nearest neighbors ($k = \min(31, \max(5, \lfloor\sqrt{n}\rfloor))$, distance-weighted, standardized); random forest (420 trees, 85% of features, bootstrap, seed 94); gradient boosting (260 trees, learning rate 0.035, depth 3, subsample 0.85, seed 95) and histogram gradient boosting (260 iterations, learning rate 0.035, 31 leaves, L2 0.05, seed 96), each with one model per output coordinate.
Development configuration	ALL ET-2, $\alpha = 1.20$; PLL ET-2, 1.20; LF ET-1, 1.00; ISL ET-4, 1.20; SSL ET-1, 1.00; ITL ET-4, 1.00; CL ET-2, 1.10.
Primary selection grid	ET-1 to ET-4 $\times\ \alpha \in \{0, 0.25, 0.40, 0.55, 0.70, 0.85, 1.00, 1.10, 1.20\}$ per class and fold, inner leave-one-subject-out; ties broken by configuration name, then α (Table S9).
SLD	Released SpineLib code (commit 62f39cf); pyACVD remeshing (subdivision 3, 5000 clusters), 10 smoothing iterations, hole filling up to 1000, `Spine(max_angle=60)`, shape decomposition without lamina; landmarks snapped to the input mesh; LPS↔RAS conversion; centerline approximated by cross-section centroids.
Software	Python 3.11, NumPy 2.4, SciPy 1.17, pandas 3.0, scikit-learn 1.8, trimesh 4.12, VTK 9.6, PyVista 0.47, pyACVD 0.4.

Sensitivity of the atlas selection to the computing environment The reported results use the ICP alignments stored with the code release, from which every number is regenerated exactly. When all alignments of the primary evaluation were recomputed from scratch in a second computing environment (same code, same pinned library versions, and the same input files, verified by checksum), the selected atlas vertebra changed for 2 of the 14 held-out vertebrae (S01 L3 and S02 L3), and the changed training-side predictions altered the inner model selection of the S01 and S02 folds. The macro-averaged MAE became 3.208 mm for GCZ and 2.418 mm for GCZ-RL (Pass@5 83.1% and 89.8%); SLD, which does not use ICP, was unchanged at 2.224 mm. The S03 fold was unchanged. Most of the difference comes from S02 L3, for which the newly selected atlas had a lower quick-ICP residual (1.33 against 1.51 mm) but a GCZ error higher by 4.0 mm, that is, an alignment with a small surface distance but wrong correspondence. The improvement of GCZ-RL over GCZ (24.6%) is preserved, whereas the ordering of GCZ-RL and SLD by macro-averaged MAE is not. Atlas selection starts ICP from the scanner coordinates of each vertebra, and several candidate alignments converge to local minima with residuals of 4–5 mm; the selection is therefore sensitive to small numerical differences.

Correction of the calibration regressor comparison The leave-one-vertebra-out comparison of Table 3 evaluates, for each held-out calibration vertebra, the corrections predicted for its GCZ points. In the version reported in the first author's dissertation, the validation rows of the bilateral classes (LF, ISL, ITL, CL) were taken as the first n rows of that vertebra and class, where n is its number of GCZ points; because original and mirrored samples alternate, half of these rows were mirrored samples, and their predicted corrections were applied to the wrong GCZ points. In addition, zones, class prior, and correction bound were computed once from all 20 calibration vertebrae. Table S6 compares the published values with the corrected protocol used in Table 3, in which only original samples are validated, each with its own GCZ point, and zones, prior, and bound are recomputed without the held-out vertebra. The correction lowers the errors of LF, ISL, and CL for every family; for ITL the corrected errors are 0.16–0.44 mm higher, mainly because zones, class prior, and correction bound no longer include the held-out vertebra. Under the corrected protocol Extra Trees is the best family for every class; the conclusion that Extra Trees is the most suitable residual regressor is unchanged. The development configuration had been fixed before the correction and is unchanged.

Table S6: Mean over classes of the leave-one-vertebra-out calibration MAE (mm) under the published and corrected protocols, and the MAE of the bilateral classes under the corrected protocol (published values in parentheses).

	Ridge	KNN	Random forest	Gradient boosting	Extra Trees
Mean, published protocol	3.103	3.017	2.754	2.777	2.575
Mean, validation rows corrected	2.986	2.655	2.241	2.274	1.984
Mean, rows and fold quantities corrected (Table 3)	3.055	2.715	2.307	2.337	2.037
LF	3.546 (3.760)	2.744 (3.653)	2.448 (3.329)	2.386 (3.543)	2.125 (3.354)
ISL	1.981 (2.013)	1.918 (1.945)	1.588 (1.831)	1.622 (1.919)	1.551 (1.805)
ITL	4.046 (3.816)	4.085 (3.833)	3.399 (3.097)	3.532 (3.090)	3.170 (3.014)
CL	4.817 (5.214)	3.700 (5.214)	2.934 (5.386)	2.963 (5.157)	2.578 (5.049)

S6. Diagnostic analysis of the correspondence rule

Pure Hungarian matching minimizes the total distance but can pair anatomically non-corresponding points, such as left–right swaps, superior–inferior inversions, and long jumps along curved attachment regions. To quantify this, an anatomy-constrained correspondence was evaluated for a geometry-only GCZ prediction on the nine held-out lumbar vertebrae of the development analysis with complete landmark sets (S02 L1–L4 and S03 L1–L5): ALL and PLL used side-aware matching with a superior–inferior band, LF and ITL strict left–right constraints, CL a four-quadrant strategy, and ISL and SSL local ordering and distance constraints; pairs that violated these rules were penalized or rejected.

Important difference from the main evaluation. In this diagnostic, which was run with the original analysis notebook, the number of GCZ landmarks per class was set to the number of reference points of the evaluated vertebra (at least four), so that nearly every reference point could be paired. This uses the cardinality, though not the positions, of the evaluated annotations. The analysis therefore characterizes the matching rules only. Its predictions are not those evaluated in the main text, which use a fixed number of at most ten landmarks per class, and its values are not comparable with Tables 4–6.

The nine vertebrae carry 1145 reference landmarks. Pure Hungarian matching paired 1126 of them (coverage 98.3%), whereas the constrained rules, which leave a landmark unpaired when no admissible candidate exists, paired 1039 (90.7%) and 1042 (91.0%). At this lower coverage, the final constrained rule had an MAE of 3.67 mm against 4.85 mm, an RMSE of 5.34 mm against 7.71 mm, a pass rate of 82.2% against 75.2%, and 58 instead of 119 errors above 10 mm (Table S7, Fig. S1). The error statistics of the three rows are computed over different sets of pairs and are not directly comparable, and part of the improvement may come from landmarks that the constrained rules leave unpaired. The analysis supports the narrower conclusion that anatomical constraints remove long-range mispairings. Table S8 reports the per-ligament MAE under the initial and final rules. The largest change was for ALL, from 5.07 to 4.17 mm, after side-aware and superior–inferior band constraints; PLL improved slightly. The other classes kept the initial rule, because further band, principal-axis, and ordering variants did not improve the balance between MAE, pass rate, and long errors.

Table S7: Pooled metrics of the diagnostic geometry-only GCZ prediction under three correspondence strategies (nine held-out lumbar vertebrae, 1145 reference landmarks). Coverage: paired reference landmarks as a percentage of all reference landmarks. Rows are computed over different sets of pairs. $N > 10$: pairs with an error above 10 mm.

Correspondence strategy	Pairs	Cov.%	MAE	SD	RMSE	Pass%	$N > 10$
Pure Euclidean Hungarian	1126	98.3	4.85	5.99	7.71	75.2	119
Initial anatomy-aware matching	1039	90.7	3.84	4.30	5.76	80.2	61
Final anatomy-constrained	1042	91.0	3.67	3.87	5.34	82.2	58

Table S8: Per-ligament MAE (mm) of the diagnostic geometry-only GCZ prediction under the initial and final correspondence rules (nine held-out lumbar vertebrae).

Lig.	Initial MAE	Final MAE	$N > 10$
ALL	5.07	4.17	13
PLL	3.70	3.62	3
LF	3.21	3.21	14
ISL	4.43	4.43	15
SSL	1.86	1.86	0
ITL	2.68	2.68	0
CL	3.36	3.36	13

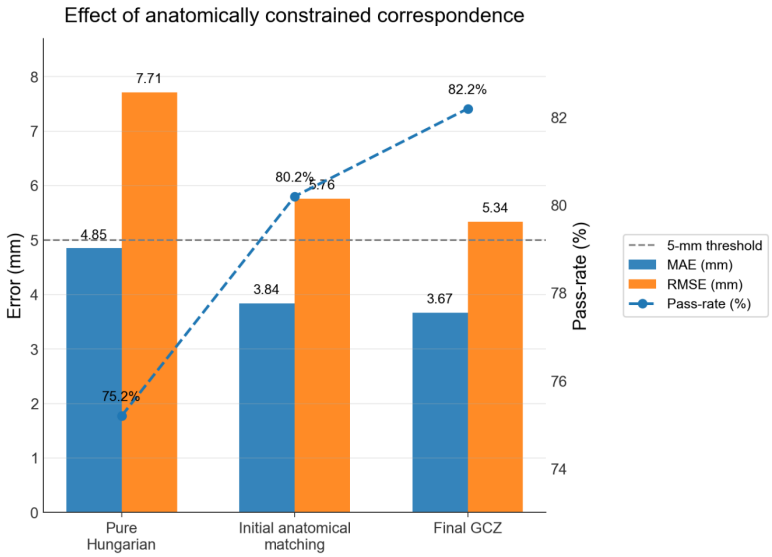


Figure S1: Effect of anatomically constrained correspondence on the diagnostic geometry-only GCZ prediction. Bars show pooled MAE and RMSE, and the dashed line shows the percentage of matched pairs within 5 mm (values of Table S7).

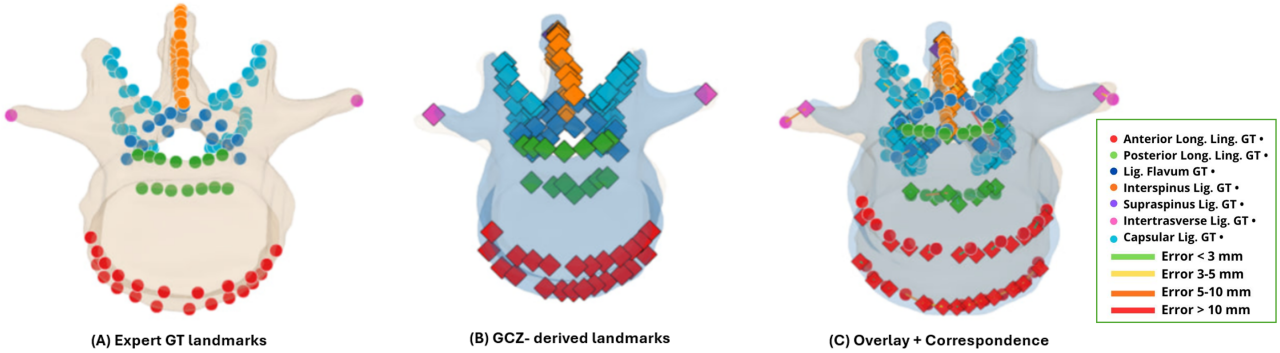


Figure S2: Representative qualitative landmark correspondence on a held-out vertebra of the diagnostic analysis. (A) Expert landmarks. (B) GCZ landmarks transferred from the atlas after alignment. (C) Overlay of expert and GCZ landmarks with correspondence lines; circles indicate expert landmarks and diamonds GCZ landmarks. The overlay exposes side swaps, superior–inferior inversions, and candidate-region failures.

S7. Models selected in the primary evaluation

When the inner validation error was identical for several configurations, as for $\alpha = 0$, where the configuration does not matter, the first configuration in name order is listed.

References

Blomenkamp, L., Kramer, I., Bauer, S., Schöche, T., 2026. SLD: Segmentation-based landmark detection for spinal ligaments, in: 2026 IEEE 23rd International Symposium on Biomedical Imaging (ISBI), pp. 1–5. https://doi.org/10.1109/ISBI61048.2026.11515825.

Table S9: Extra Trees configuration and blend weight selected for each class by the inner leave-one-subject-out of each outer fold, with the inner validation MAE (mm). $\alpha = 0$ keeps the GCZ point.

	Held-out S01		Held-out S02		Held-out S03	
Class	Model, α	MAE	Model, α	MAE	Model, α	MAE
ALL	ET-4, 0.55	3.234	ET-3, 0.25	3.886	ET-1, 0	3.163
PLL	ET-1, 0	2.188	ET-1, 0	3.427	ET-1, 0	1.823
LF	ET-1, 0	4.482	ET-1, 0.40	1.873	ET-1, 0	2.195
ISL	ET-3, 0.70	4.603	ET-2, 0.70	2.325	ET-4, 0.85	2.106
SSL	ET-1, 0.25	2.745	ET-2, 0.40	3.309	ET-1, 0.25	3.516
ITL	ET-2, 1.20	2.218	ET-1, 1.20	1.694	ET-4, 1.20	1.130
CL	ET-2, 0.25	4.780	ET-1, 0	2.882	ET-1, 1.20	2.379